What is a property worth?

Predicting bina.az prices with a decision tree and an SVM written from scratch

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The problem

Given a listing from bina.az, we ask two questions.

A • How much is it worth?

Regression on $\log(1 + \text{price})$.

Metrics: RMSE, MAE, R^2

B • Premium or standard?

Premium means price above the *training* median, 208 000 AZN.

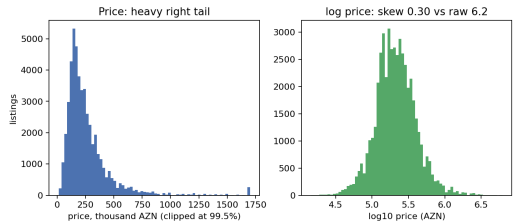
Metrics: F1, ROC-AUC, confusion matrix

57 480	27	70/15/15	8 622
clean listings	features	train / val / test	test listings

Cleaning the data

Step	Rows
Raw Kaggle dump	100 775
Latest scrape per ad	63 773
Residential only	57 796
Sanity bounds (area, price/m^2)	57 712
Duplicate content removed	57 480

- The same ad is scraped daily, so we keep the latest.
- Missing values are structural: houses have no floors, flats have no land.



Price is heavy-tailed, log price is close to symmetric.

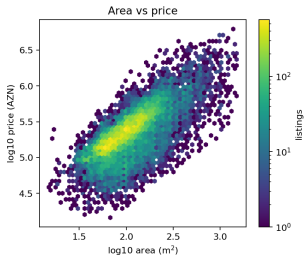
Leakage

`unit_price` and `total_price` restate the target: correlation with price is **1.0000**. We drop them, together with identifiers, free text and post-listing signals (views, VIP).

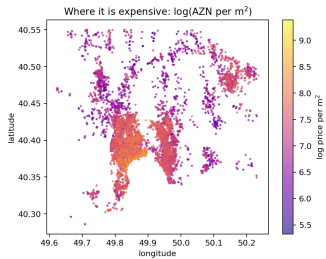
What keeps the rest honest:

- Imputation, scaling and the 208 000 AZN threshold are fitted on train only.
- District encoding and neighbour prices are computed out-of-fold.
- The split is grouped by physical flat, so re-posted copies never straddle it.
- The test set is used once, for the final numbers.

Size and location



Area against price.



Price per m^2 by location.

Area drives price, but price per m^2 changes sharply across the city. So each listing also gets the mean and spread of log price per m^2 of its 20 nearest training listings (out-of-fold, so a row never sees its own label).

Decision tree

$$G = 1 - \sum_k p_k^2 \qquad I_{\text{reg}} = \frac{1}{N}\sum_i (y_i - \bar{y})^2$$

$$\text{IG} = I(\text{parent}) - \sum_c \frac{N_c}{N} I(c)$$

- Sort each feature, test the midpoints, keep the best gain, recurse.
- Stop on depth, leaf size, split size or minimum gain.

Final: regression depth 10, leaf 15; classification Gini, depth 10, leaf 30.

Validation R^2, regression (leaf 15)

depth	6	8	**10**	12
R^2	0.812	0.835	**0.840**	0.838

Too shallow underfits, deeper starts to overfit.

Validation AUC, classification

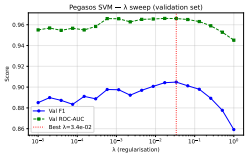
Gini, depth 10, leaf 15	0.9565
Entropy, depth 10, leaf 15	0.9574
Gini, depth 10, leaf 30	**0.9600**

Pegasos SVM

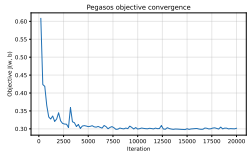
$$J(w, b) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{n}\sum_i \max\bigl(0, 1 - y_i(w^\top x_i + b)\bigr)$$

$$w \leftarrow (1 - \eta_t \lambda)\, w + \eta_t\, y_i x_i\, [\, y_i w^\top x_i < 1\,], \quad \eta_t = \frac{1}{\lambda t}$$

- $\|w\| = 1.67$, margin width 1.20, 34% of training points are support vectors.
- Random Fourier features (RBF) did not beat linear: F1 0.85 vs. 0.91.



Validation F1 against λ.



Objective during training.

How we compared

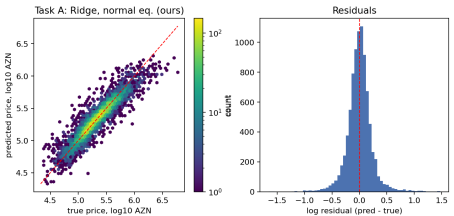
- Same split for every model: 70/15/15, grouped by flat, seed 42.
- Hyperparameters chosen on validation only.
- One evaluation module, tested against scikit-learn, scores every model.
- Uncertainty: bootstrap std over 300 resamples, plus nested 5-fold CV.
- Baselines: mean and majority floors, ridge, scikit-learn tree, `SVC`, `SGDClassifier`.

Results: price

Model	RMSE	R^2	MAE (AZN)
Mean (floor)	0.629	0.000	127 850
Ridge (ours)	**0.226**	**0.870**	**48 280**
Tree (ours)	0.239	0.856	49 516
sklearn tree	0.238	0.856	49 353

Test set, RMSE and R^2 on log price.

Ridge beats the tree: once location is encoded well, price is close to linear.

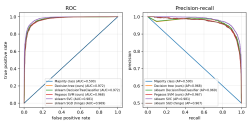


Predicted vs. true price.

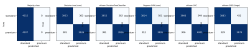
Results: premium or standard

Model	F1	AUC	Fit
Tree (ours)	0.914	0.972	65 s
sklearn tree	0.914	0.972	0.7 s
Pegasos (ours)	**0.908**	**0.968**	**1 s**
sklearn SVC	0.924	0.981	306 s
sklearn SGD	0.913	0.969	0.1 s

- Our tree matches scikit-learn.
- Pegasos is 1.6 F1 points behind SVC and about 300× faster.



ROC and PR curves.



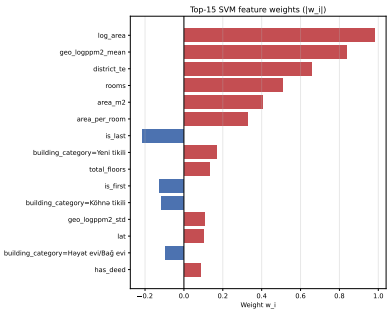
Confusion matrix.

What drives the price

Tree importance	Task A	Task B
Area (m^2)	0.616	0.648
Neighbourhood price (kNN)	0.211	0.086
District encoding	0.123	0.187
Everything else	0.050	0.079

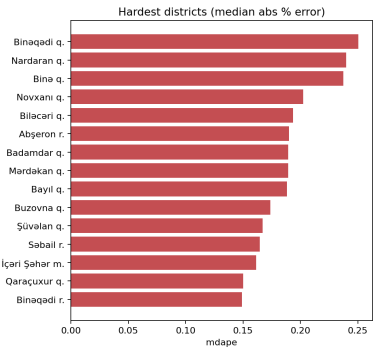
Area and location carry 92 to 95% of the importance.

The SVM agrees: its largest weights are log-area, neighbourhood price and district encoding.



Largest SVM weights.

Where it fails



Hardest districts, median error.

- Above 800k AZN the median error is 32%, with a −27% bias.
- Houses are hardest: 22.7% error vs. about 10% for flats.
- The 20 biggest misses are all under-predictions, mostly large houses in Badamdar, Merdekan and Bilgeh.
- Task B errors sit near the 208k threshold.

Limits and next steps

Limits

- Asking prices, not sale prices.
- 27 tabular features, no text or photos.
- Luxury listings are rare and under-predicted.
- Premium is relative to the training median.

Next

- Land-area features for houses.
- Features from listing descriptions.
- A quantile model for the luxury tail.
- A random forest from our own trees.

Team and reproducibility

Emil	cleaning, leakage-safe preprocessing, evaluation, ridge, pipeline, tests
Gasham	decision tree from scratch, tree experiments, scikit-learn comparison
Orkhan	Pegasos SVM and random Fourier features, SVM experiments, comparison

Everything reproduces with one command: `python -m src.run_all --strict`

https://github.com/Emilsdeyta/ml-final-project-bina-az

Thank you. Questions?